

Two-way fixed effects estimators with heterogeneous treatment effects*

Clément de Chaisemartin[†] Xavier D’Haultfoeulle[‡]

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Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

Keywords: linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

JEL Codes: C21, C23

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[†]University of California at Santa Barbara, clementdechaisemartin@ucsb.edu

[‡]CREST-ENSAE, xavier.dhaultfoeulle@ensae.fr

$t - 1$ and t would have experienced if they had remained treated. In staggered adoption designs, Assumption 9 and 10 are not necessary for identification, because no group leaves the treatment. Together, Assumptions 5 and 10 imply that the ATE follows the same evolution over time in every group: $E(\Delta_{g,t}) = \eta_t + \theta_g$.¹³ This still allows for heterogeneous treatment effects across groups and over time.¹⁴

Assumption 11 (*Existence of “stable” groups*) For all $t \in \{2, \dots, T\}$:

1. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 0$ and $D_{g,t} = 1$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 0$.
2. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 1, D_{g,t} = 0$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 1$.

The first point of the stable groups assumption requires that between each pair of consecutive time periods, if there is a “joiner” (i.e., a group switching from being untreated to treated), then there should be another group that is untreated at both dates. The second point requires that between each pair of consecutive time periods, if there is a “leaver” (i.e., a group switching from being treated to untreated), then there should be another group that is treated at both dates.

Notice that under Assumption 11, groups’ treatments are not independent, so Assumption 3 cannot hold. Accordingly, we replace Assumption 3 by Assumption 12 below. Assumption 12 requires that conditional on its own treatments, a group’s outcomes be mean independent of the other groups’ treatments. It is weaker than Assumption 3. Assumption 11 is necessary to show that our estimator is unbiased, but it is not necessary to show that it is consistent. Accordingly, in Section ?? of the Web Appendix, we show that our estimator is consistent under Assumption 3. For every $g \in \{1, \dots, G\}$, let $\mathbf{D}_g = (D_{1,g}, \dots, D_{T,g})$.

Assumption 12 (*Mean independence between a group’s outcome and other groups treatments*) For all g and t , $E(Y_{g,t}(0)|\mathbf{D}) = E(Y_{g,t}(0)|\mathbf{D}_g)$ and $E(Y_{g,t}(1)|\mathbf{D}) = E(Y_{g,t}(1)|\mathbf{D}_g)$.

We can now define our estimator. For all $t \in \{2, \dots, T\}$ and for all $(d, d') \in \{0, 1\}^2$, let

$$N_{d,d',t} = \sum_{g: D_{g,t}=d, D_{g,t-1}=d'} N_{g,t} \quad (3)$$

¹³ It should be possible to weaken Assumptions 9-10, in particular to account for dynamic effects where $\Delta_{g,t}$ may depend on $(D_{g,1}, \dots, D_{g,t-1})$. This would however introduce complications that are beyond the scope of this paper.

¹⁴ Imposing Assumptions 9 and 10 does not change the decompositions obtained in Theorems 1 and 2. $Y_{g,t}(1)$ is observed for all the treated (g, t) cells entering these decompositions, so those assumptions do not bring identifying information for those cells.